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Qualitative and Quantitative Study of Wound Healing Processes in the Coelenterate, *Plexaurella fusifera:* Spatial, Temporal, and Environmental (Light Attenuation) Influences

Adriana Meszaros and Charles Bigger

Department of Biological Sciences, Florida International University, University Park Campus, Miami, Florida 33199

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Following injury, the Caribbean soft coral, *Plexaurella fusifera,* forms an epithelial front containing amoebocytes and zooxanthellae, a photosynthetic endosymbiont. Amoebocytes may be responsible for extruding the connective mesogleal fibers necessary for regeneration of tissue and zooxanthellae may provide the energy for repair. This study examined the effects of time, space, and environment (light attenuation) on wound healing in this coral species and quantitatively confirmed the increase of amoebocyte concentrations in the injured area. A wound was made on coral branchlets by removing ~4.5 mm of coenenchyme. At assigned times after injury, samples were collected for gross morphological and histological evaluation, in which amoebocytes and zooxanthellae concentrations were quantified within 0.009 mm^3 of tissue. Overall amoebocyte numbers within uninjured and wounded tissue were similar. However, when numbers of amoebocytes per area of injured tissue were calculated and compared to those of uninjured tissue, 82.4% more amoebocytes occurred at distances 0–0.5 mm from the wound edge, while areas of tissue >2 mm from the wound edge were occupied by fewer amoebocytes. Overall increases in concentrations of zooxanthellae also occurred within wounded coral, but no apparent temporal, spatial, or light-related pattern was detected. Therefore, this study supports the conjecture that amoebocyte accumulation at a wound site is an effect of cells migrating from uninjured tissue adjacent to the wounded edge. In addition, this movement occurs regardless of light attenuation. Light, which in this study was confined to ranges between 70 and 545 μE s^{-1} m^{-2}, did not significantly affect the wound healing process in regard to either closure or cellular concentrations. © 1999 Academic Press

***Key Words: Plexaurella fusifera;* wound healing; inflammation; gorgonian; cellular concentrations; amoebocyte; zooxanthellae; light attenuation.**

INTRODUCTION

Gorgonians, horny corals, are colonial coelenterates. Under normal conditions, the tissue is often damaged by predation or mechanical injury, e.g., as a result of storms. In some areas, this has been further exacerbated by anthropogenic effects, e.g., damage due to anchors, sightseer handling or standing on the colonies, collecting, and vessel groundings.

As in all metazoans, the integrity of the tissue constitutes a first line of defense against pathogens. The coenenchyme, the colonial tissue between the individual polyps, has an epidermis covering a collagenous matrix populated by amoebocytes (Chester, 1913; Bigger and Hildemann, 1982). In a phylogenetic context, the first blood cells may have been derived from these mobile phagocytic cells (Pieter *et al.,* 1993). Interspersed within this matrix, or mesoglea, are canals lined with gastrodermal cells whose vacuoles house symbiotic algae, also known as zooxanthellae (Barnes and Chalker, 1992).

Recognizing and responding to wounds with a nonspecific immunological mechanism (Lackie, 1980), coelenterates have the ability to regenerate lost tissue (Mattson, 1976). Following wounding, amoebocytes, the putative immunocytes of the anthozoans (Hildemann *et al.,* 1977) stream into the area of damage (van Praet and Doumenc, 1975), but evidently do not proliferate in the wound area (Young, 1974; Patterson and Landolt, 1979). In fact, amoebocytes are not normally found undergoing mitosis (Chapman, 1974). Therefore, this local reaction of living tissue to injury (Boyd, 1990) is an inflammatory response and seems to be unique to coelenterates within the classes Anthozoa and Schyphozoa (Metchnikoff, 1892; Patterson and Landolt, 1979; Bigger and Hildemann, 1982). A wound causing total coenenchymal loss seems to be a sink for amoebocytes, with the source of cells being unknown. Three proposed sources are: (1) proliferation at the site, (2) differentiation from stem cells or dedifferentiation from other cell

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types (Chapman, 1974; Lang da Silveira and Van't Hof, 1977; Rinkevich, 1996), and/or (3) migration from organized storage sites or from adjacent tissue (Young, 1974).

Gorgonians are believed to possess a permanent and independent reserve of energy, which is available for future healing processes (Wahle, 1983). Since amoebocytes lack a known nutritive role (Bigger and Hildemann, 1982), a potential source of usable energy is thought to be available from the photosynthetically active algal symbionts (zooxanthellae). When corals are wounded, zooxanthellae can sometimes be visualized outside the gastrodermis (McKinstry *et al.*, 1989), indicating a disruption of the confining endodermal cell membranes, the activation of an expulsion process, or possibly recruitment. If gorgonian metabolism, which is dependent to a degree on zooxanthellae, is jeopardized, then high energy functions, such as wound healing and regeneration, could be affected.

The main objective of this study was to perform wound healing studies on the coelenterate *Plexaurella fusifera* to describe the processes leading to repair. Histologically obtained cell count data were gathered to quantify changes in major cell populations (amoebocytes and zooxanthellae) during wound healing. Temporal, spatial, and environmental (light attenuation) effects on cellular concentrations were analyzed.

It was not within the scope of this study to distinguish between mesogleal amoebocytic types, but rather to describe these cells as a group at the wound site. We determined that amoebocyte accumulation at the wound site was influenced by spatial factors, whereby migration of amoebocytes from surrounding tissue caused a depletion of cell number in adjacent tissue. In addition, overall increases in zooxanthellae occurred within wounded coral, but no apparent temporal, spatial, or environmental pattern was noted.

MATERIALS AND METHODS

Collection and Preparation

Soft coral was collected at Tavernier, FL, 150–200 m offshore in 3 m of water. Branchlets, 15 cm in length, were removed with scissors from five *P. fusifera* colonies, identified according to spicules after transport (Bayer, 1961). Samples were transported with aeration in perforated baskets placed in a cooler full of *in situ* sea water. Samples were acclimated and transferred into 20-L aquaria. Each aquarium was setup with aerators, underwater gravel filters, and artificial sea water (Tropical Marine Salt). Light was supplied by two 250-W metal halide bulbs 0.8 m above the water surface. Automatic timers maintained a 14/10 light/dark cycle. Colonies were fed commercial invertebrate formula. Salinity was maintained between 34 and 36 ppt.

Each branchlet was cut into three pieces, 4.5 cm long. The remaining branchlet was used in spicule analysis. Ten millimeters of tissue was scraped with a scalpel from the end of each piece and the bare skeleton was placed into an apparatus that secured the samples 15–20 cm below the water surface. Samples were allowed to acclimate for 2 weeks: 1 week at a range of 465–545 $\mu E\ s^{-1}\ m^{-2}$ light exposure and a 2nd week at the designated light treatment levels. The light level reaching the sample was measured using a Biospherical Model QSL-100 light meter.

Procedure

A wound was made by removing 4.25–4.75 mm of coenenchyme 1.5 cm from the end of each coral sample using a ring template and a scalpel. During histological preparation, all samples were fixed in Bouin's overnight, ethanol dehydrated, cleared in xylene, embedded in paraffin at 56°C (Paraplast Regular; Sigma, St. Louis, MO), and sectioned (American Optical Model 820 rotary microtome). Twenty-four 6-µm-thick longitudinal sections were serially taken from each block just beyond the skeletal axis. Six sections were stained for each of the following four stains (total 24 sections): (1) Gill's Hematoxylin and Eosin Y (H&E), (2) Feulgen/Fast Green, (3) Periodic Acid Schiff/Hematoxylin, or (4) Mallory-Heidenhain.

Aquaria temperatures were recorded every minute and averaged every hour using thermoprobes (Type T Gauge 24 thermocouple wire) attached to a Campbell Scientific Inc., 21× micrologger. The difference in temperature among tanks on any one day reached a maximum of 3°C. The light intensity in aquaria without shading was measured at 465–545 $\mu E\ s^{-1}\ m^{-2}$. Using neutral density filters, other aquaria were exposed to 195–235 and 70–120 $\mu E\ s^{-1}\ m^{-2}$ levels of light. These levels of light are comparable to normal light and shade environments (Dubinsky *et al.*, 1984; Porter *et al.*, 1984). Seventy-two pieces were prepared from each of the five colonies. Twenty-four pieces were placed into each of the three separate light exposures. For each light exposure, 12 of these pieces were wounded and 1 piece was fixed at different times after wounding (initial, 6 h, 12 h, 24 h, each day subsequently for 6 days, 14 days, and 21 days). The remaining 12 pieces were not wounded and 1 piece was fixed to correspond to each wounded piece being fixed. Photographs of gross pathology were taken at each time point. Degree of closure was noted as the shortest distance between axial fronts and was measured either with a manual caliper or with an ocular micometer, depending on spatial constraints.

Cell Counts

Two cell types were targeted for quantification: amoebocytes and zooxanthellae. Amoebocytes were found in

the mesoglea (Chester, 1913) and were PAS positive (bright field microscopy). A stain series inclusive of glycogen degradation, carbohydrate test with acetylation, glycogen stain, low/mixed/high Diamine stain, and Alcian blue was performed in order to confirm PAS positivity (Troyer, 1980; Clark, 1981). Zooxanthellae were found in the gastrodermis and were easily visible as green fluorescent cells in PAS-stained sections (axioscope filter 15 with ultraviolet microscopy). Cells were counted in a calibrated (600-µm-diameter) field of view (20× objective) on a compound microscope. Each sectioned sample was considered as quarters (horizontal line along the skeletal axis and vertical line through the middle of the section in uninjured controls or through the center of the wound area). One quarter was picked randomly for quantification, with the exception that quarters with polyps in view were avoided due to their innately high levels of zooxanthellae (Yonge and Nicholls, 1931; Zvalinskii *et al.*, 1980). Six distinct cell-count regions were quantified for each quarter (Fig. 1). Region 0s was defined as the area, or field of view, that included 0.0–0.5 mm of the cut edge at the epidermal surface, s (0–1 mm inward from the epidermis to the axis). Region 2s was defined as the surface area, s, 1.5–2 mm from the wound. Region 4s was the surface area, s, 3.5–4 mm from the wound. Regions 0i–4i corresponded to regions 0s–4s, respectively, except that these regions were at the skeletal axis designated the interior, i, tissue (1–2 mm inward from the epidermis to the axis). Amoebocyte and zooxanthellae number in each region were tallied using a cell counter. Each of the six regions was counted in five sequential quarters and added. Therefore, counts are represented as cell number/0.0085 mm^3 of tissue. In addition, since the five colonies had innately varying levels of both amoebocytes ($P = 0.0001$) and zooxanthellae ($P = 0.0272$), wounded cellular concentrations are adjusted for uninjured colonial cellular concentrations and are reported as wounded minus uninjured concentrations. Cell counts were subjected to unpaired t tests or analyses of variance and differences among means were determined by Scheffe F-test (Statview 512+, 1986; Abacus Concepts, Inc., Calabasas, CA).

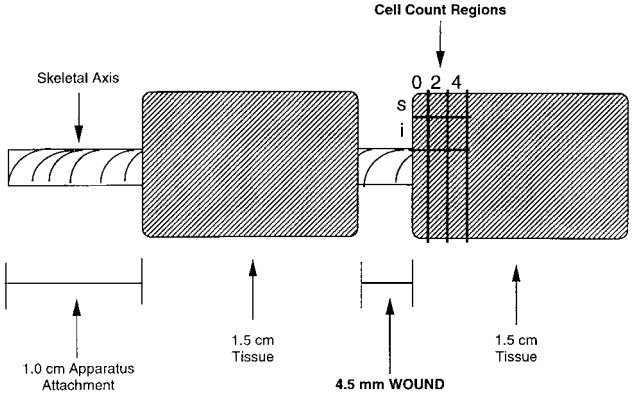

FIG. 1. Schematic representation of placement and size of wound. Cell count regions 0–4s and 0–4i are also represented. Numbers 0, 2, and 4 designate three distances 0, 2, and 4 mm from the edge of the wound in wounded samples and from a random point in uninjured (normal) samples. The letters s and i designate two depths, surface and interior, which represent 0–1 and 1–2 mm from the epidermis toward the axis.

RESULTS

Normal Morphology and Cytology

Amoebocytes. See Fig. 2A. The amoebocytes associated with wound healing were amoeboid shaped, often elongated, and ranged in size from 5 to 9 µm in width and from 8 to 15 µm in length. All of these cells were PAS positive. The nuclei were irregularly round to almost oval and had diameters of 3.5 to 4 µm. They were strongly basophilic positive and contained vacuoles that made them appear like epithelial cell nuclei. The cytoplasm was filled with small to medium granules (0.5 to 1.5 µm in diameter). These granules were mostly eosinophilic positive. However, some amoebocytes had slightly basophilic granules. A variety of histochemical tests, listed under Materials and Methods, were performed to confirm PAS positivity (data not shown).

Zooxanthellae. See Fig. 2B. These endosymbiotic dinoflagellates occurred within vacuolated gastrodermal cells. They stained differentially from yellow to blue. The nucleus of the symbiotic gastrodermal cell appeared flattened against the periphery of the cell.

Pathology of Wound Healing

The initial wound left an interstice of 4.25–4.75 mm between the two opposing healing fronts. The distance between the healing fronts diminished over time until the fronts fused and no interstice was visible. The healing rate ranged from 0.32 to 0.42 mm/day from the 4th day to the end of the 2nd week of wound repair.

The wound caused tissue disorganization and consisted of two opposing cut edges of tissue held together by the remaining axis. These edges were at a 90° angle to the axis. Four major stages were noted during the 3-week healing period: sealing, repair, fusion, and reorganization. All times given express periods postwounding.

During the 1st day, tissue was sealed from the environment by epithelial tissue that extended downward toward the axis. The epithelium next to the wound became flatter. In addition, a flat extension of cells spread over the skeletal axis, forming a layer approximately 50 µm thick (Fig. 3A). This layer of tissue, composed mainly of amoebocytes (Fig. 3B), was designated the axial front. The area between the axial

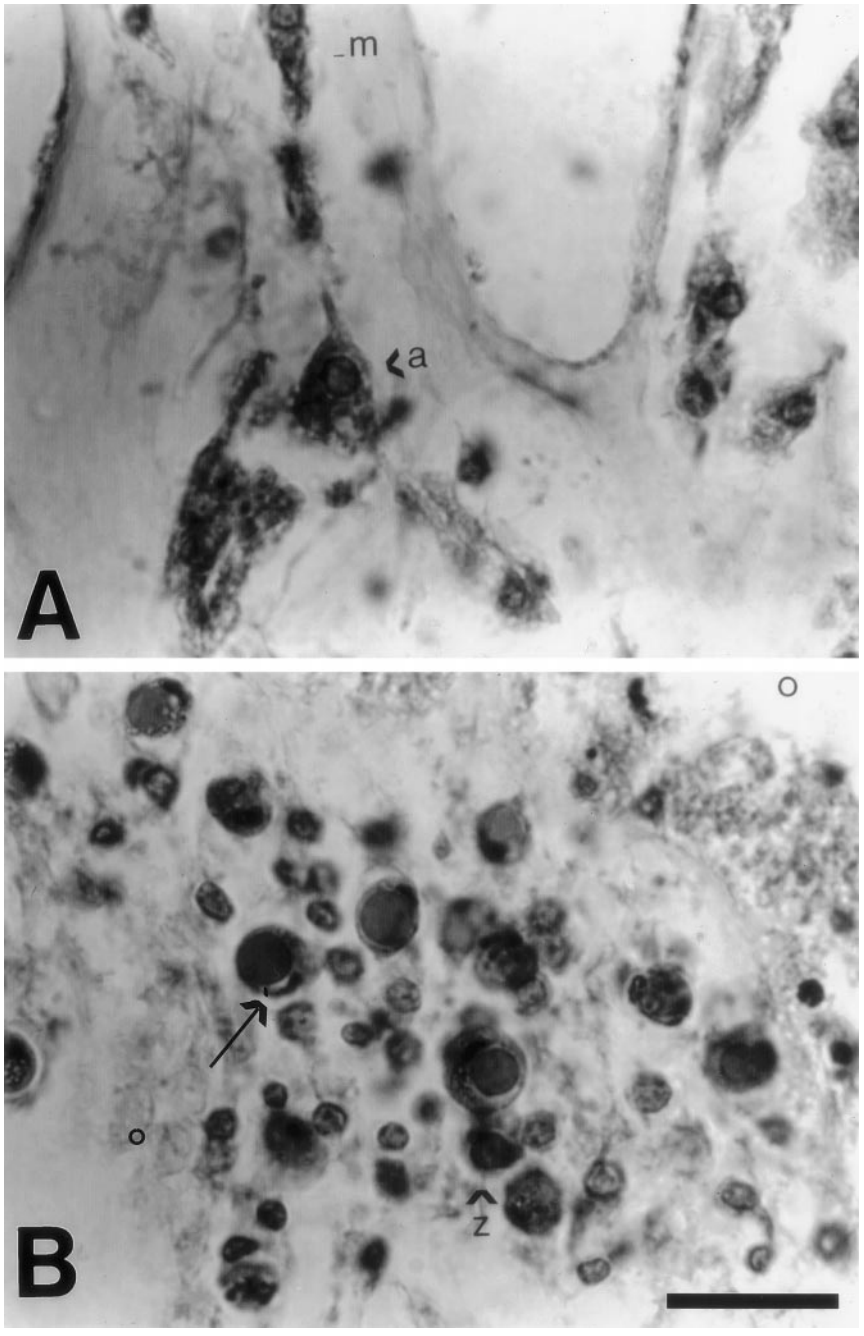

FIG. 2. Bright field micrographs of normal cytology. (A) Amoebocytes (a) are very abundant within the cell cords of the mesoglea (m). The amoebocyte shown represents a subpopulation often found at the injury area. (H&E). (B) Zooxanthellae (z), photosynthetic endosymbionts, are dinoflagellates found within the vacuoles of gastrodermal cells. The elongated nuclei of the gastrodermal cells are seen along the edge of the cell (arrow). Solenia (o). (H&E). Bar, 16 μM.

front and the extending epithelial tissue was sealed mostly by amoebocytes and was designated the healing front. These amoebocytes, whose axes were perpendicular to the healing front, migrated into the area from adjacent tissue (Fig. 3C). Amoebocytes are capable of sending out processes that anastomose with each other (Batham *et al.*, 1961). The lining on the outer edge of the healing front seemed to be formed by such a flattening and joining of amoebocytes (Fig. 3D).

Repair, which began after the formation of the axial and healing fronts, proceeded until the end of the 1st week. The axial fronts thickened radially outward from the axis to the epithelial edge and the healing front thickened (Figs. 4A and 4B). Amoebocytes amassed not

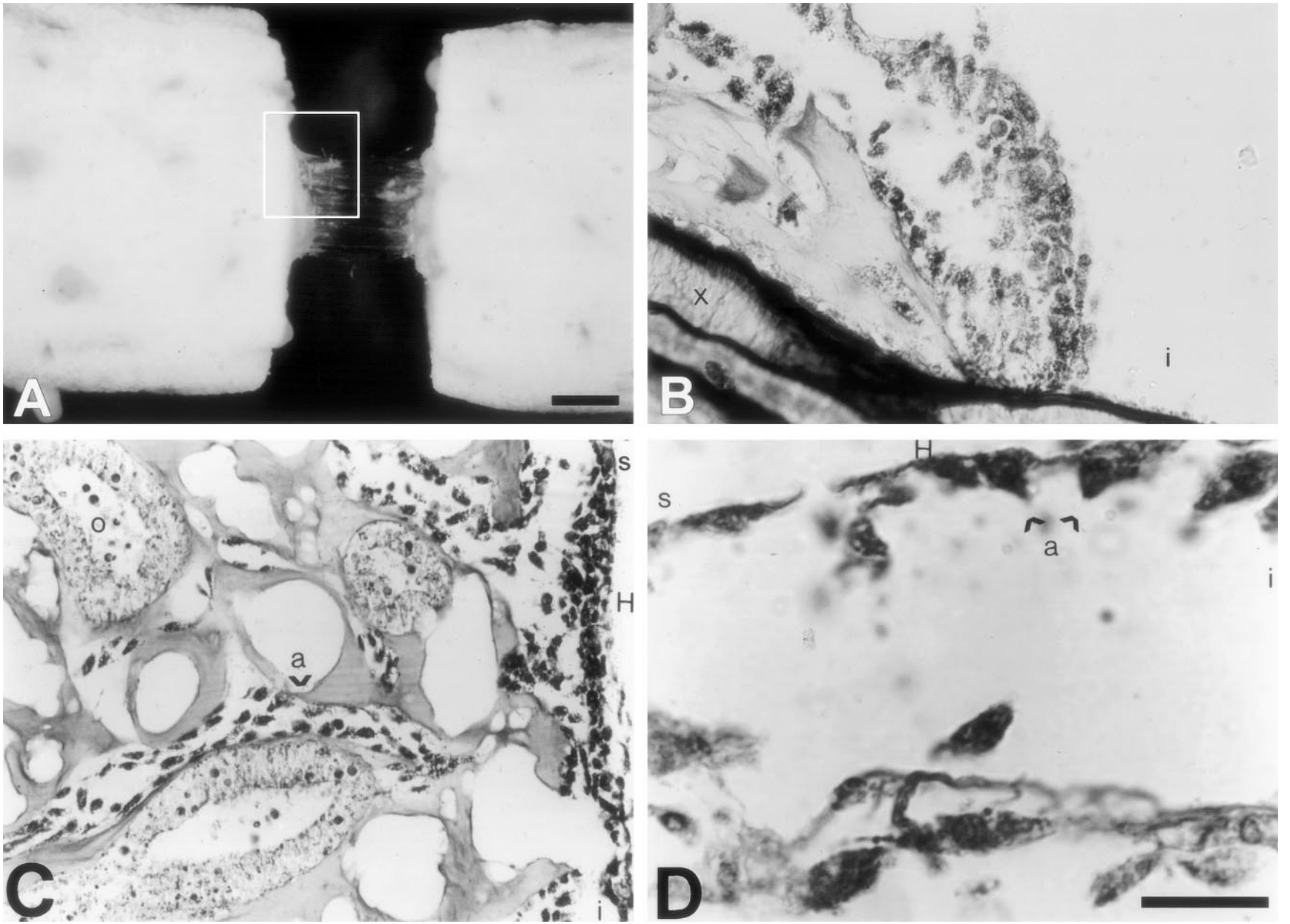

FIG. 3. The pathology of wound healing (stage 1). (A) Gross pathology of a 0.5-day wound with tissue spreading from the wound edge (white box). Bar, 1.25 mm. (B) The layer of tissue (white box magnified) at the interior of the wound edge (i) extends over the skeletal axis (x) to form the axial front. (Mallory). Bar (in D), 32 µM. (C) Amoebocytes (a) accumulating in the edge of the healing front (H) from adjacent tissue. Interior of wound (i); surface of wound (s); solenia (o). (Feulgen). Bar (in D), 65 µM. (D) The healing front (H) between the surface (s) and the interior (i) of the wound seemed to be formed by a flattening and joining of amoebocytes (a). (Mallory). Bar, 16 µM.

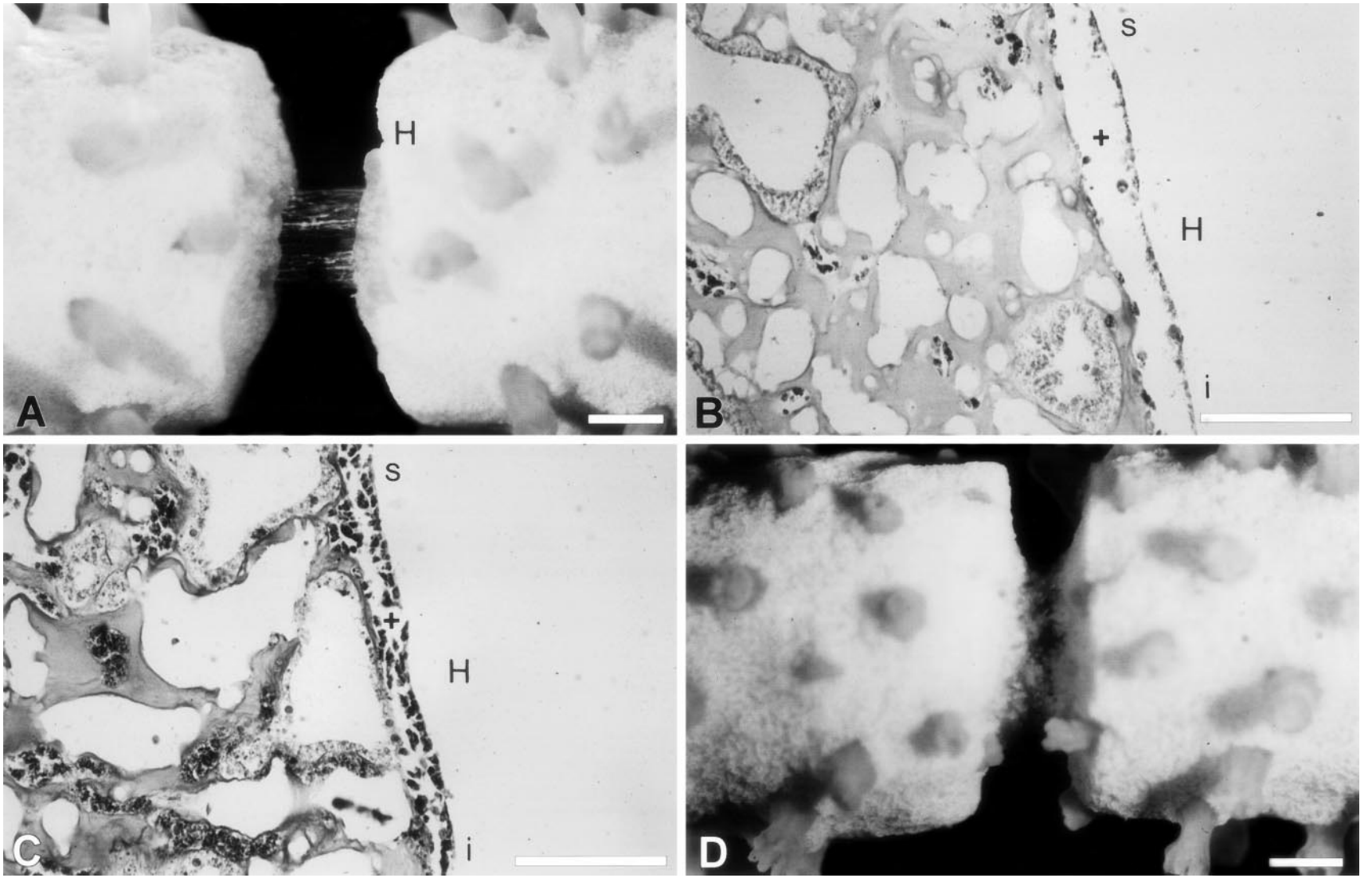

FIG. 4. The pathology of wound healing (stages 2 and 3). (A) Gross pathology of a 4-day wound showing the progression of the opposing healing fronts (H) along the skeletal axis. Bar, 1.25 mm. (B) The healing front (H) between the surface (s) and the interior (i) of the wound during the early repair stages was delineated from the cut edge by an area composed of few cells (+). (Mallory). Bar, 125 μM. (C) The healing front (H) between the surface (s) and the interior (i) of the wound thickened and amoebocytes amassed in this area (+ compared to + in B) during the late stages of repair. (Mallory). Bar, 125 μM. (D) Gross pathology of a 7-day wound showing stage 3, fusion, of wound healing pathology. Bar, 1.25 mm.

only within the axial front but also within the healing front. Eventually, the axial front radially thickened to the point at which it was no longer discernible from the healing front (Fig. 4C compared to Fig. 4B).

The opposing fronts of the wound progressed at equal healing rates (0.15–0.23 mm/day) and fused by 1 week (Fig. 4D). The surviving tissue adjacent to the wound did not have a stretched appearance, but appeared less condensed than normal tissue. During the next 2 weeks, the surface portions of the fronts extended over the area of fusion and the tissue in the area of fusion became more organized. The only evidence of the wound after 3 weeks of healing was a slight indentation.

Cell Counts

Amoebocytes and zooxanthellae were counted in specimens from three colonies which had been either wounded or uninjured (normal controls) and treated to different light intensities. Cells were tabulated at different times (day 1, 4, 7, and 14) in six areas at or near the wound area (Fig. 1) and at corresponding times and areas at or near the projected middle of the uninjured sections.

First, differences in cellular concentrations inherent in uninjured specimens were determined. Examined using an ANOVA, uninjured amoebocyte concentrations were not affected by light ($P = 0.4103$), time ($P = 0.2897$), or space ($P = 0.2091$). On the other hand, uninjured zooxanthellae concentrations were affected by light ($P = 0.0068$) and space ($P = 0.0001$), but not time ($P = 0.6551$). Zooxanthellae levels were highest in samples exposed to 195–235 $\mu E\ s^{-1}\ m^{-2}$ light and in all interior tissue areas. These inherent differences would not have accounted for differences in wounded versus uninjured samples.

Cell counts in uninjured and wounded specimens were compared to determine involvement of wounding on cellular concentrations. Overall numbers of amoebocytes were not significantly different between wounded and uninjured samples. However, the median number of amoebocytes counted in wounded samples (993.5) was notably different from uninjured samples (944.5) (Fig. 5A). Significant increases in numbers of zooxanthellae were noted in wounded samples as a whole compared to uninjured samples ($P = 0.0001$) (Fig. 5B).

Amoebocyte Concentrations

Increases in amoebocyte numbers in wounded samples were spatially influenced ($P = 0.0001$). Neither environmental (light) ($P = 0.2632$) nor temporal ($P = 0.0604$) factors significantly affected amoebocyte concentrations. Although not statistically significant, a peak in amoebocyte accumulation was noted at day 4.

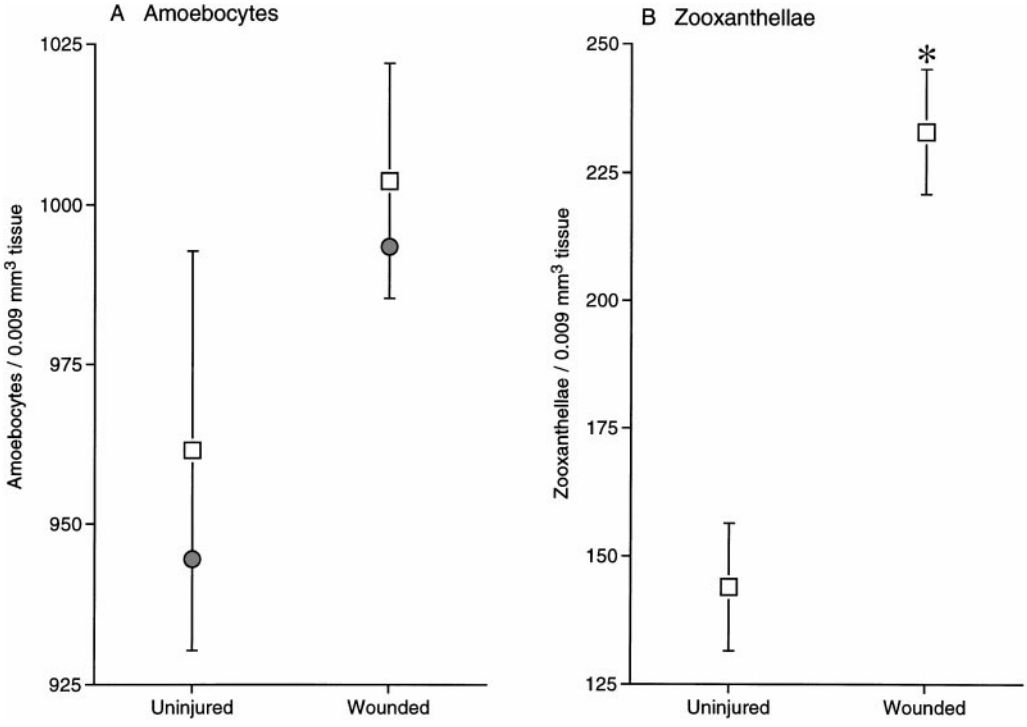

FIG. 5. Scatterplots of amoebocyte (A) and zooxanthellae (B) mean cell count/0.009 mm^3 of uninjured and wounded tissue. Lines represent standard error for 108 counts. Filled circles in (A) represent the median values for the observations. $*P < 0.05$ for the difference between wounded and uninjured cell counts.

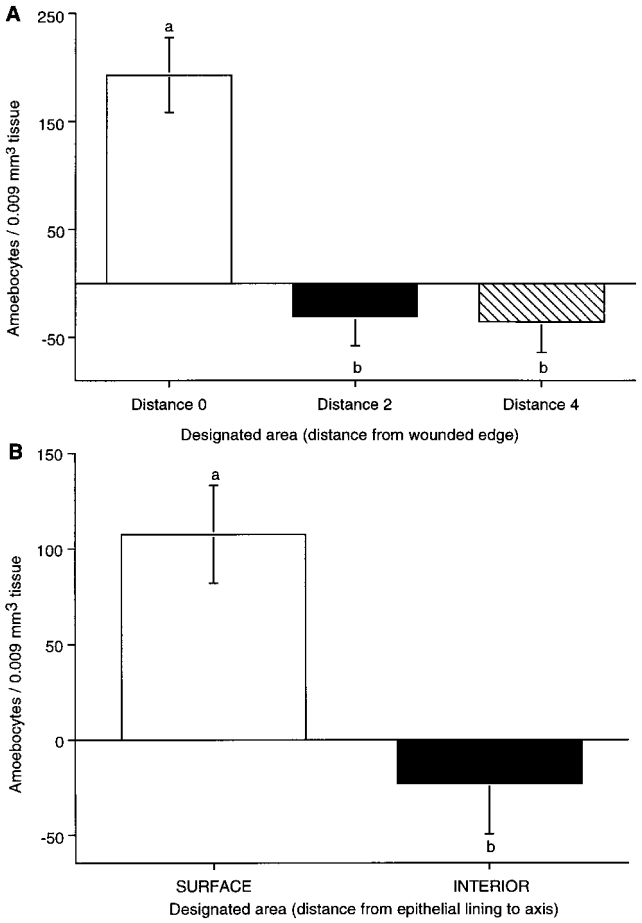

FIG. 6. Amoebocytes/0.009 mm^3 of tissue (A) at 0.0–0.5 mm (distance 0), 1.5–2.0 mm (distance 2), and 3.5–4.0 mm (distance 4) from the wound edge and (B) at 0–1 mm (surface) and 1–2 mm (interior) from the epithelial lining toward the skeletal axis. Bars represent standard errors for 72 (A) and 108 (B) counts. The counts plotted are wounded minus uninjured counts. Values labeled with the same letter are not different ($P \geq 0.05$) according to Scheffe's F test.

Spatially, amoebocyte numbers increased to levels above normal at the wound front and decreased to levels below normal at the areas adjacent to the wound front (Fig. 6A). These decreases were concentrated in interior tissue areas (Fig. 6B).

Zooxanthellae Concentrations

Increases in zooxanthellae numbers in wounded samples were spatially ($P = 0.0001$) and temporally ($P = 0.0038$) influenced. No significant differences ($P = 0.1133$) were found between zooxanthellae numbers in the wounded samples under different light regimes. However, concentrations of zooxanthellae in wounded samples were consistently higher than those in unwounded samples, regardless of light. Zooxanthellae numbers differed between surface and interior tissue areas. This pattern, however, is inherent to normal samples as well (Fig. 7A). There was no significant difference in zooxanthellae concentration between regions close to and those displaced from the wound front ($P = 0.2649$). Therefore, the increase in concentration was uniform over the distance measured from the wound (Fig. 7B). The zooxanthellae numbers in wounded samples peaked at day 7 (Fig. 7C), which coincides with the energy-expensive fusion period of healing.

DISCUSSION

Temperatures in this study were not outside the temperature range normally encountered by the coral in the collection area and signs of thermal stress, such as pigmentation redistribution or tissue sloughing (Jokiel and Coles, 1977; Gates *et al.*, 1992), were not observed. Temperature was not considered a major variable in this study.

Healing rates (0.30–0.45 mm/day) remained constant throughout the 2-week period. Closure time was comparable to that of *Plexaura flexuosa* (Lang da Silveira and Van't Hof, 1977) and *Swiftia exserta* (Olano, 1993), which lacks zooxanthellae. The light attenuation defined by this study did not affect the rate at which wounded tissue obtained closure.

The cytology, gross pathology, and cytological events of repair in *P. fusifera* were similar to those of cells and events described in sea anemone (Chapman, 1953; Batham, 1961; Young, 1974), *Plexaura homomalla* (Bayer, 1974), *P. flexuosa* (Lang da Silveira and Van't Hof, 1977), and *S. exserta* (Olano, 1993). As with all the above gorgonians, the amoebocytes of *P. fusifera* were arranged into cell cords ("strands") within the mesoglea. Fusion of the axial fronts occurred within the 1st week of healing. The epidermis surrounding the wound moved across the cut surface and amoebocytes migrated from the mesoglea into the wound.

The tissue directly at the wound edge (within 0–1 mm) consistently had more amoebocytes than the surviving tissue adjacent to the wound edge (within 1–4 mm). Within uninjured specimens, the amoebocyte concentration was homogeneous throughout the 4-mm-wide region. Therefore, the increase of amoebocyte levels in the tissue at the wound edge was concluded to be an effect of the wound healing processes. These data coincide with observations made by a variety of authors (Metchnikoff, 1892; Young, 1974; Lang da Silveira and Van't Hof, 1977; Patterson and Landolt, 1977; Olano, 1993) that amoebocyte levels increase in an area of injury. Data presented here quantitatively confirm this increase.

The origin of the excess of amoebocytes seen at the wound area was previously unknown. The increase of amoebocytes at the wounded edge was assumed to have arisen by one or a combination of the following three mechanisms: (1) proliferation at the site, (2) differentiation from stem cells or dedifferentiation from other cell

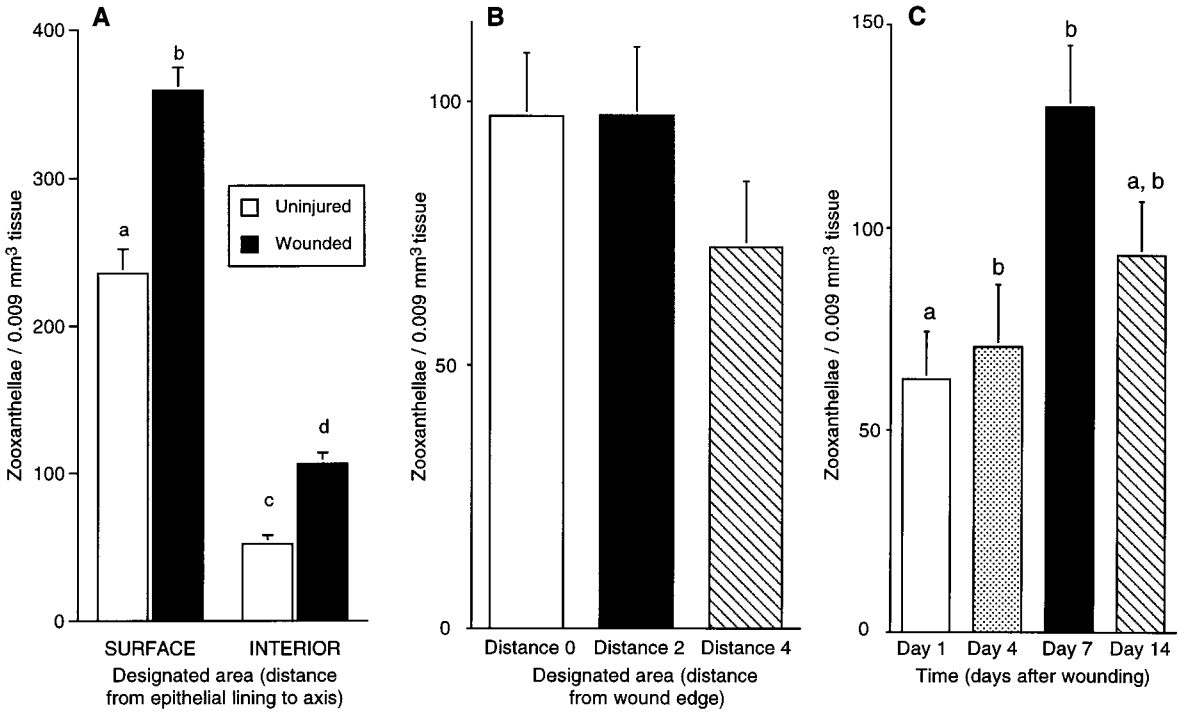

FIG. 7. Zooxanthellae/0.009 mm^3 of tissue (A) at 0–1 mm (surface) and 1–2 mm (interior) from the epithelial lining toward the skeletal axis; (B) at 0.0–0.5 mm (distance 0), 1.5–2.0 mm (distance 2), and 3.5–4.0 mm (distance 4) from the wound edge; and (C) at day 1, 4, 7, and 14 after wounding. The counts plotted in (B and C) are wounded minus uninjured counts. Bars represent standard errors for 52–108 counts. Values labeled with the same letter are not different ($P \geq 0.05$) according to Scheffe's F test.

types (Chapman, 1974; Lang da Silveira and Van't Hof, 1977; Rinkevich, 1996), and/or (3) migration from organized storage sites or from adjacent tissue (Young, 1974).

Since different subgroups of amoebocytes were not classified and the method for viewing mitotic figures was not exact, the occurrence of proliferation could have been overlooked in this study. However, evidence indicating that accumulation was not due to proliferation was threefold. First, no mitotic increase has been observed in the mesoglea after wounding (Young, 1974). Second, amoebocytes decreased in concentration adjacent to the wounded edge and were oriented toward the wound edge, indicating movement rather than proliferation. Third, the wound healing process did not cause an overall increase in amoebocyte concentration in the region within 0–4 mm adjacent to the injury compared to uninjured samples. Rather, the distribution of amoebocytes was spatially rearranged within the 4-mm-wide region, indicating a redistribution rather than an increase in numbers.

A drop in amoebocyte number was noted between days 4 and 7, which coincided with the reformation of a normal epithelial layer at the surface portions of the wound edge. Therefore, amoebocytes, the putative stem cells in Anthozoa (Chapman, 1974), may have been differentiating or dedifferentiating into other cells types, such as epithelial cells, between days 4 and 7. These changes in amoebocyte concentrations due to their function as stem cells would impact the later repair stage of wound healing.

The origin of the amoebocytes within the wounded edge during the sealing stage of wound healing is unknown. However, since a decrease of amoebocyte numbers compared to uninjured specimens did occur in tissue 1–4 mm from the wounded edge, the increase of amoebocytes in the area of injury was assumed to be a function of migration from tissue located up to 4 mm away from the wound edge. Olano (1993) described a "stretching" of adjacent tissue due to a net movement of coenenchymal tissue into the wound area. Conversely, in *P. fusifera*, no gross net movement of tissue was observed. However, this species has more tissue surrounding the central axis and the larger diameter may have camouflaged any tissue movement.

Data presented here indicate that mechanisms explaining changes in amoebocyte concentrations at the wound edge are not mutually exclusive. Migration of amoebocytes from organized storage sites or from adjacent tissue definitely characterize the initial stages of wound healing, e.g., sealing. However, other mechanisms, such as stem cell differentiation, may play a role during the later stages of wound healing, e.g., repair.

The alterations in light levels did not affect amoebocyte numbers in uninjured coral. In addition, the increased numbers of amoebocytes adjacent to the

injury within wounded samples was not attributable to environmental (light) alteration. Amoebocytes are not known to be photosensitive and, therefore, were not expected to react to changes in light.

Changing light levels did not affect the rate at which the coral closed the wound. In addition, varying light intensities did not explain the higher numbers of zooxanthellae observed in wounded coral compared to uninjured coral, and this quantity of zooxanthellae was not affected by light attenuation. Due to the complexity of the experimental design and the analytical power limitations of the statistical tests employed, more subtle effects of light may not have been detected. For instance, the effect of wounding seemed to overshadow the effect that varying light levels may have on steady state concentrations of zooxanthellae.

Nonetheless, uninjured corals maintained at a light intensity of 70–120 $\mu E\ s^{-1}\ m^{-2}$ had fewer zooxanthellae than corals maintained at light intensities of 195–545 $\mu E\ s^{-1}\ m^{-2}$. This study, therefore, does not support the hypothesis that the coral's zooxanthellae concentration does not change with changing light intensities. However, the investigations on which that postulation was based examined a different subclass of corals and included observations of cellular concentration within polyps (Yonge and Nicholls, 1931; Porter *et al.*, 1984), whereas this study examined coenchymal tissues. In addition, the wounded surface tissue layer had higher concentrations of zooxanthellae than those of the wounded interior tissue layer. This phenomenon was also common in uninjured coral specimens. Since they are autotrophs, zooxanthellae are photosensitive. It was assumed that more light penetrated the surface layers of the tissue than the interior layers and, therefore, more zooxanthellae were found in the higher light intensities at the surface.

The function of increasing concentration of zooxanthellae in wound healing activities is unknown and could not be determined by this study. Zooxanthellae were initially assumed to play a role in the energetics of wound healing due to their role in coral metabolism (Falkowski, 1990). However, the rate of healing within *P. fusifera* was comparable to the rate of healing within *S. exserta*, which lacked zooxanthellae. In addition, the increase of zooxanthellae numbers was consistent throughout the tissue examined, rather than increasing predominantly at the wounded edge. These observations invalidate the supposition that the increase in zooxanthellae concentration in wounded coral is caused solely by the increased demand for energy in the repair zone or by a movement of gastrodermal canals. The following ideas could be considered: (1) areas adjacent to injury, which also play a role in the repair process, have increased energy requirements resulting in an increased zooxanthellae population, (2) repair processes may have disrupted a mechanism used by corals to regulate the concentration of zooxanthellae within their tissue, or (3) gastrodermal cells are proliferating and coincidentally increasing zooxanthellae numbers.

The present study has demonstrated that *P. fusifera* has injury repair processes and cytology similar to those described in other gorgonians. For the first time, this study of *P. fusifera* quantitatively confirmed the proposed increase of amoebocyte concentrations in an injured area and supported the conjecture that amoebocytes migrate from uninjured tissue adjacent to a wounded edge during the early stages of repair. Further studies, perhaps tracking specific amoebocytes, should be conducted to determine the origin of the accumulating amoebocytes. As discussed above, zooxanthellae concentrations increased throughout the tissue up to 4 mm from the wounded edge and, therefore, zooxanthellae should be considered somehow part of the wound healing processes. Further studies involving "blocking" energy translocation between the coral tissue and the zooxanthellae, such as chemical inhibitors or darkness, should be conducted to determine the reason for such concentration increases.

Light, which in this study was confined to normal ranges, was not found to have a significant effect on the wound healing process in regard to either closure or cellular concentrations. This conclusion does not rule out the possible synergistic effects of environmental variables found in nature. Further study of environmental effects on wound healing could reveal some of these relationships. For instance, commercial pollutants have already been shown to significantly affect the wound healing processes in other coelenterates (Adams and Kennedy, 1992).

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